Gemma3

> **Demo Environment**

Development Board: Jetson Orin series motherboard

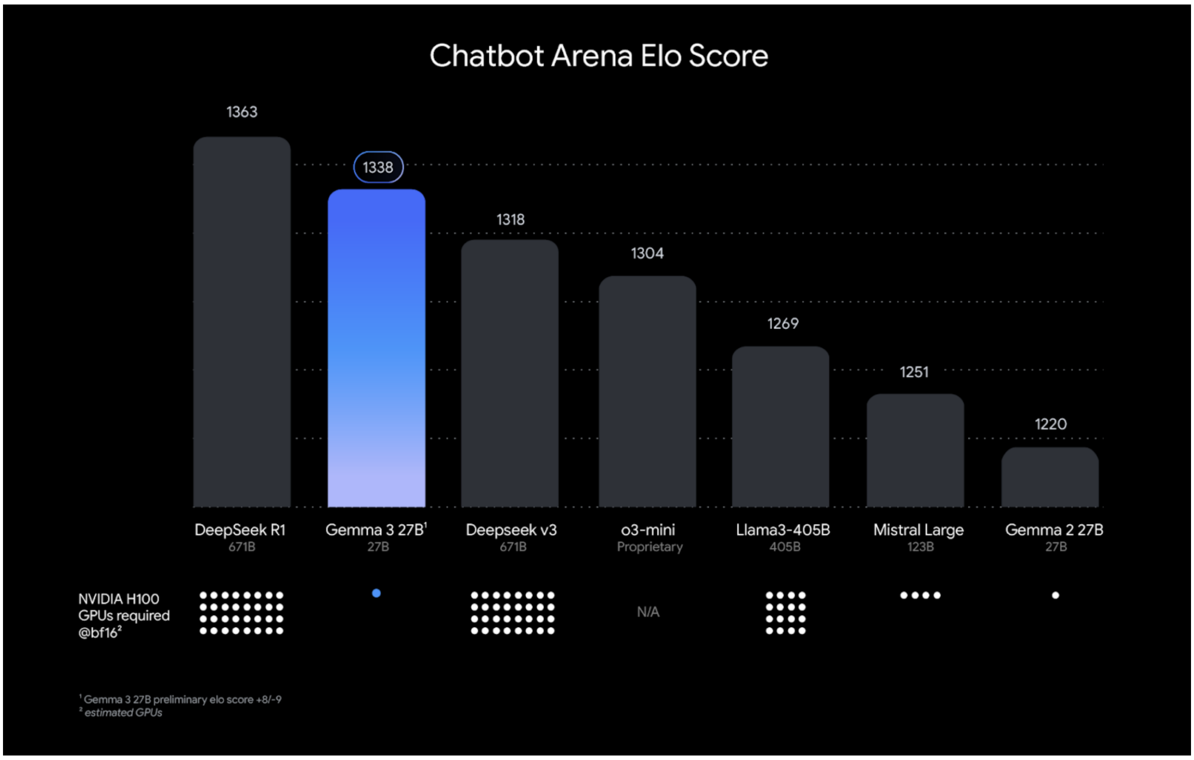
Note: Due to performance limitations, the Jetson Orin Nano 4GB requires the reduced-parameter version

Gemma is a family of lightweight models built by Google based on Gemini technology. The Gemma 3 model is multimodal (capable of processing text and images), has a 128KB context window, and supports over 140 languages.

1. Model Size

Model	Size
gemma3:1b	815MB
gemma3:4b	3.3GB

2. Performance

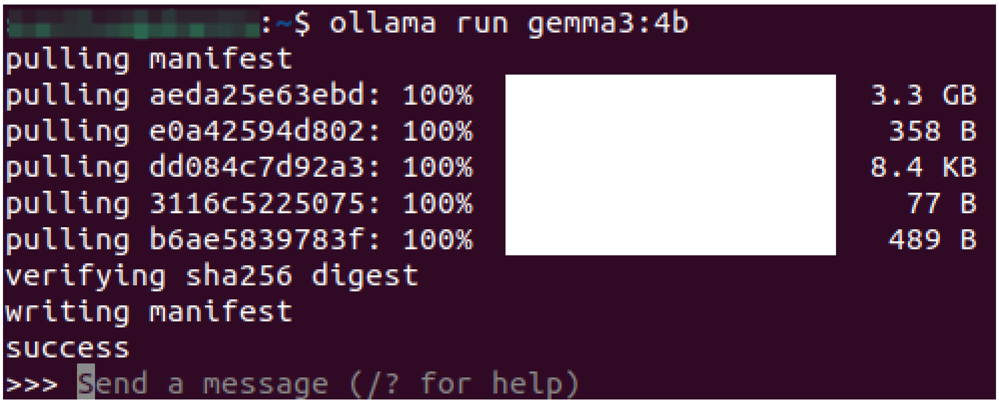


3. Using Gemma3

3.1 Running Gemma3

Use the run command to run the model. If the model is not already downloaded, it will automatically pull the model from the Ollama model library:

```
ollama run gemma3:4b
```



3.2 Start a conversation

```
How to learn a programming language?
```

Response time depends on hardware configuration, so please be patient!

```
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>>> How to learn a programming language?
Learning a programming language can feel daunting at first, but with a
structured approach and consistent effort, it's absolutely achievable.
Here's a breakdown of how to learn a programming language effectively,
broken down into stages and with different learning styles in mind:

**1. Choosing Your Language:**

* **Beginner-Friendly Options:**
    * **Python:**  Highly recommended for beginners. It has a clean
syntax, huge community support, and is used in many fields (web
development, data science, scripting).
    * **JavaScript:**  Essential for web development (front-end and
increasingly back-end with Node.js).
    * **Scratch:** (Visual, block-based) A fantastic starting point for
kids and absolute beginners to understand programming concepts without
worrying about syntax.
* **Consider Your Goals:** What do you want to *do* with programming?
    * **Web Development:** JavaScript, Python (with frameworks like Django
or Flask), Ruby
    * **Data Science/Machine Learning:** Python, R
    * **Game Development:** C#, C++, Python (with Pygame)
    * **Mobile App Development:** Swift (iOS), Kotlin/Java (Android)
```

3.3 Visual Function

```
What do you see in this picture? :./test_pic.png
#Using ": + the image path" in the conversation allows the model to use its
visual function and interpret the information in the image.
```

```
[redacted]:~$ ollama run gemma3:4b
>>> What do you see in this picture?:./test_pic.png
Added image './test_pic.png'
Here's a breakdown of what I see in the picture:

*   **Main Subject:** A young man is in the foreground, with his eyes
closed and his head tilted back, seemingly taking a deep breath. He's
wearing a white tank top and shorts.
*   **Background:** The background is a cityscape, likely a rooftop with
buildings visible. There is a water bottle next to him.
*   **Other Person:** There’s another man sitting on the right, also
seemingly relaxed.
*   **Sky:** The sky is a vibrant blue with fluffy white clouds.
*   **Color Tone:** The image has a slightly warm, saturated color tone,
giving it a dramatic feel.

This photograph evokes a sense of stillness, perhaps a moment of respite
amidst a busy urban environment. It's a strong, visually compelling image.

>>> Send a message (/? for help)
```

3.4 Ending the Conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

References

> **Ollama**

Official Website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

> Gemma3

Ollama Compatible Model: https://ollama.com/library/gemma3